
CBE 30355 Transport I Midterm Exam

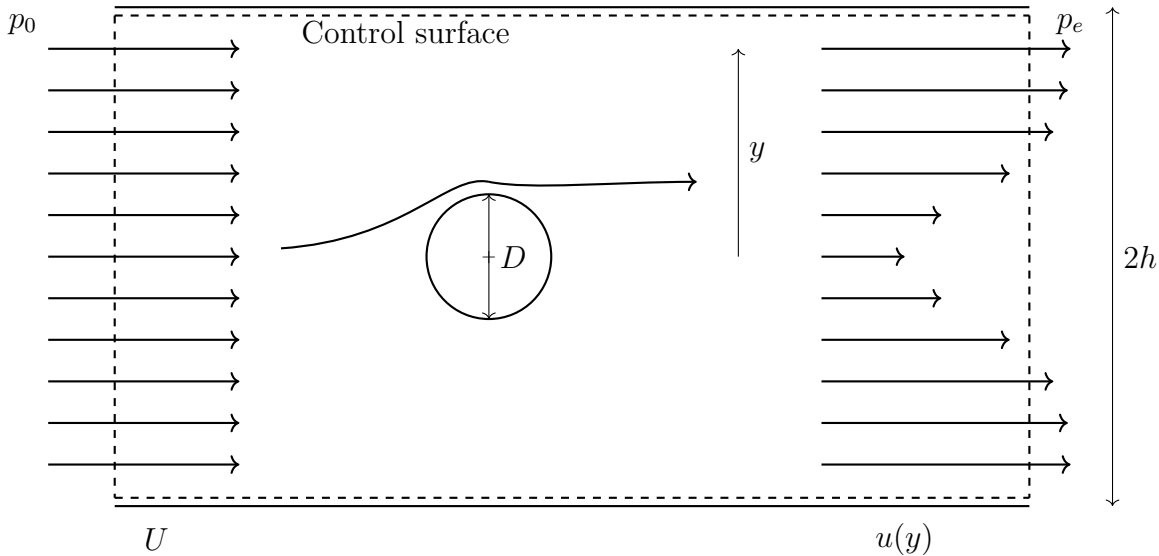
October 16, 2025 Closed Books and Notes

Problem 1

(20 points)

Momentum Balance in Wake Traverse Experiment

In Senior Laboratory one of the experiments used to be measuring the drag on a cylinder at high Reynolds number. One of the techniques used is called a *wake traverse*, in which we measured both the uniform velocity U upstream of the cylinder, and the non-uniform velocity $u(y)$ downstream of the cylinder. We also measured the upstream pressure p_0 and the downstream pressure p_e . The diagram from the lab manual is given below.



- Using conservation of mass (or volume) write down the relationship between the upstream velocity and the integral of the downstream velocity.
- Write down in words the integral momentum balance around the control surface depicted above.
- Given that we can neglect any drag on the side walls of the control surface, develop an explicit integral expression for the drag per unit length (into the paper) F/L exerted by the fluid on the cylinder in terms of p_0 , p_e , U , and $u(y)$. Both p_0 and p_e may be taken to be constant over the inlet and outlet, respectively.

Hint: This problem is pretty much identical to the sluice gate problem you solved for homework, except here the upstream and downstream pressures are measured (no gravity), and the downstream velocity profile is non-uniform, and thus must be left in integral form in the momentum balance.

Problem 2

(20 points)

Flow Resistance in Membranes: Unidirectional Flows

A microfiltration membrane of thickness h is often modeled as an array of cylindrical pores. The membrane is further characterized by a porosity (fraction open area) of ϵ . A common problem is that the pores are not all of the same size, which can greatly affect the flow rate. We are interested in calculating the permeability K of such a membrane, defined as the ratio of the superficial flow rate Q/A (flow rate per unit area, encompassing both the pores and solid) to the pressure drop Δp normalized by the viscosity, e.g., $Q/A = K \Delta p / \mu$.

- Starting from the Navier-Stokes equations, calculate the flow rate through a single pore of radius a .
- If the pores are of two sizes with radii a_1 and a_2 , and each occupy the same surface area (e.g., a whole lot more of the small pores), develop an expression for the permeability.
- If one of the pores has twice the radius of the other, what fraction of the total flow goes through the larger size pores?

The following equations may be helpful:

$$\frac{1}{r} \frac{\partial (ru_r)}{\partial r} + \frac{1}{r} \frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0$$

$$\rho \left(\frac{\partial u_z}{\partial t} + u_r \frac{\partial u_z}{\partial r} + \frac{u_\theta}{r} \frac{\partial u_z}{\partial \theta} + u_z \frac{\partial u_z}{\partial z} \right) = -\frac{\partial p}{\partial z} + \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u_z}{\partial \theta^2} + \frac{\partial^2 u_z}{\partial z^2} \right] + \rho g_z$$

Problem 3

(20 points)

The Clepsydra of the Ancient Greeks

Earlier this semester you (and your favorite AI) analyzed the Persian Water Clock, an example of an inflow clepsydra. The ancient Greeks used these too to limit speaking time in their assemblies - but theirs was even simpler, just a bowl with a hole in the bottom. When the water ran out, so did your speaking time! Here we analyze such an outflow clepsydra.

You can analyze any geometry of bowl, but some are much easier to do than others! Here we consider a conical bowl with an internal cone angle of 60° . For this geometry the height of water h in the cone and the volume V are simply related by $V = \frac{\pi h^3}{9}$. Our conical bowl has a small hole in the bottom with radius R_0 .

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- a. Develop the differential equation governing the change in height of water in the cone as a function of time. Use inertial scalings, and account for losses by a discharge coefficient λ .
 - b. If the initial volume in the cone is V_0 , render the problem dimensionless and via scaling determine the characteristic drainage time t_c . Note that it is probably easiest if you use V_0 to get H_0 , the initial height, using the relation above!
 - c. Solve for the dimensionless drainage time t_d^* .
 - d. If the initial volume is 1 liter and the discharge coefficient $\lambda = 0.6$, how small would the hole have to be to limit speaking time to 5 minutes?
 - e. For these conditions, comment (quantitatively) on whether the inertial approximation made above is valid.